

- 1.5** Define *physical change* and *chemical change*. State which type of change occurs in each of the following statements:
- (a) Passing an electric current through molten magnesium chloride yields molten magnesium and gaseous chlorine.
 - (b) The iron in discarded automobiles slowly forms reddish brown, crumbly rust.
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- 1.6** Which of the following is a chemical change? Explain your reasoning: (a) boiling canned soup; (b) toasting a slice of bread; (c) chopping a log; (d) burning a log.

- 1.15** Why is a quantitative observation more useful than a non-quantitative one? Which of the following are quantitative?
- (a) The Sun rises in the east.
 - (b) A person weighs one-sixth as much on the Moon as on Earth.
 - (c) Ice floats on water.
 - (d) A hand pump cannot draw water from a well more than 34 ft deep.

- 1.19** Write the conversion factor(s) for
- (a) in^2 to m^2
 - (b) km^2 to cm^2
 - (c) mi/h to m/s
 - (d) lb/ft^3 to g/cm^3

- 1.23** For each of the following cases, state whether the density of the object increases, decreases, or remains the same:
- (a) A sample of chlorine gas is compressed.
 - (b) A lead weight is carried up a high mountain.
 - (c) A sample of water is frozen.
 - (d) An iron bar is cooled.
 - (e) A diamond is submerged in water.

- 1.26** The average radius of a molecule of lysozyme, an enzyme in tears, is 1430 pm. What is its radius in nanometers (nm)?

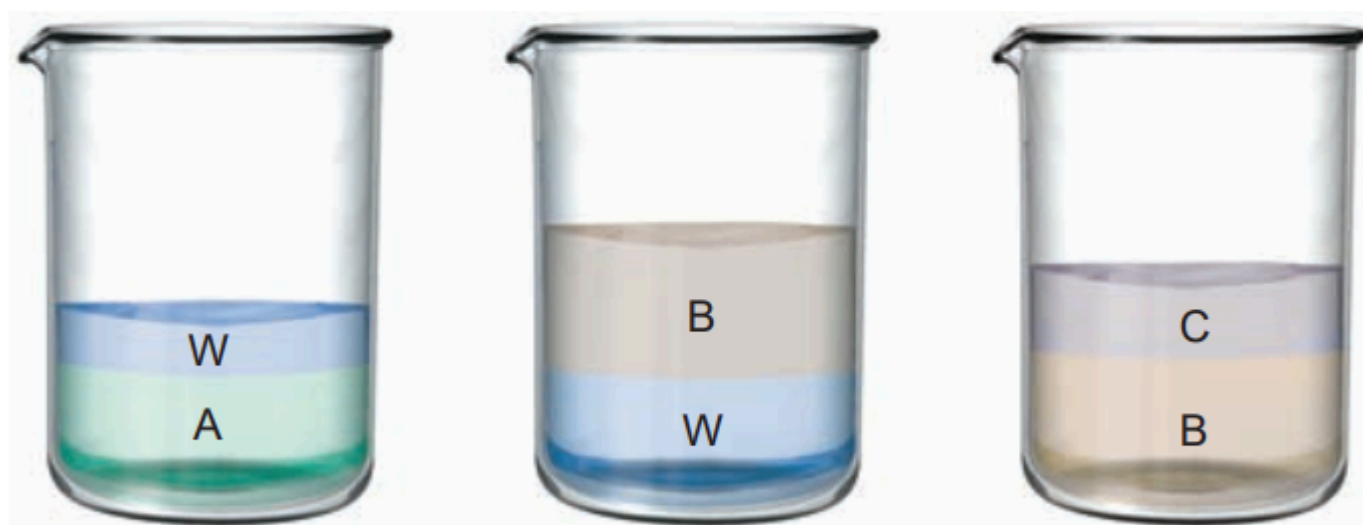
- 1.28** What is the length in inches (in) of a 100.-m soccer field?
- 1.29** The center on your basketball team is 6 ft 10 in tall. How tall is the player in millimeters (mm)?
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- 1.30** A small hole in the wing of a space shuttle requires a 20.7-cm^2 patch. (a) What is the patch's area in square kilometers (km^2)? (b) If the patching material costs NASA $\$3.25/\text{in}^2$, what is the cost of the patch?
- 1.31** The area of a telescope lens is 7903 mm^2 . (a) What is the area in square feet (ft^2)? (b) If it takes a technician 45 s to polish 135 mm^2 , how long does it take her to polish the entire lens?
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- 1.32** Express your body weight in kilograms (kg).
- 1.42** Perform the following conversions:
- (a) 68 F (a pleasant spring day) to C and K
 - (b) 164 C (the boiling point of methane, the main component of natural gas) to K and F
 - (c) 0 K (absolute zero, theoretically the coldest possible temperature) to C and F

1.45 The distance between two adjacent peaks on a wave is called the *wavelength*.

(a) The wavelength of a beam of ultraviolet light is 247 nanometers (nm). What is its wavelength in meters?

(b) The wavelength of a beam of red light is 6760 pm. What is its wavelength in angstroms (Å)?

1.46 Each of the beakers depicted below contains two liquids that do not dissolve in each other. Three of the liquids are designated A, B, and C, and water is designated W.



(a) Which of the liquids is(are) more dense than water and which less dense?

(b) If the densities of W, C, and A are 1.0 g/mL, 0.88 g/mL, and 1.4 g/mL, respectively, which of the following densities is possible for liquid B: 0.79 g/mL, 0.86 g/mL, 0.94 g/mL, 1.2 g/mL?

1.58 Carry out the following calculations, making sure that your answer has the correct number of significant figures:

(a)
$$\frac{2.795 \text{ m} \times 3.10 \text{ m}}{6.48 \text{ m}}$$

(b) $V = \frac{4}{3}\pi r^3$, where $r = 17.282 \text{ mm}$

(c) $1.110 \text{ cm} + 17.3 \text{ cm} + 108.2 \text{ cm} + 316 \text{ cm}$

1.60 Write the following numbers in scientific notation:

- (a) 131,000.0 (b) 0.00047 (c) 210,006 (d) 2160.5

1.62 Write the following numbers in standard notation. Use a terminal decimal point when needed:

- (a) 5.55×10^{-3} (b) 1.0070×10^{-4} (c) 8.85×10^{-7} (d) 3.004×10^{-3}